

Frobenius–Witt cotangent complex for derived rings

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Abstract

Shimada recently showed that Frobenius–Witt cotangent complex, the animation of Saito’s Frobenius–Witt differentials, vanishes on perfectoid rings, thus it serves as a candidate of “absolute” cotangent complex.

In this article, we give a pullback description of their arithmetic extension, and as a consequence, we give a direct description of Frobenius–Witt cotangent complex, which leads to generalizations to derived rings and animated pre-log rings. This description allows us to compute Frobenius–Witt cotangent complex of derived δ -rings as well.

We also propose a version of Frobenius–Witt cotangent complex relative to derived δ -rings, and establish a vanishing result for prisms, which generalizes vanishing of Frobenius–Witt cotangent complex of perfectoid rings.

Finally, independently of previous developments, we give a regularity criterion via Frobenius–Witt cotangent complex for p -local Noetherian (non-local) rings without F -finiteness.

1 Introduction

The ring $\mathbb{Z}$ of integers is the initial object in the category of rings, thus it could be understood as an absolute base in algebraic geometry. However, we remark that the affine scheme $\mathrm{Spec}(\mathbb{Z})$ is of dimension 1, not only in the sense of Krull dimension, but also cohomologically. For our purposes, we know that the cotangent complex $L_{\mathbb{F}_p/\mathbb{Z}}$ does not vanish, and for a commutative $\mathbb{F}_p$ -algebra R , say, the non-vanishing of $L_{\mathbb{F}_p/\mathbb{Z}}$ leads to junk data in the cotangent complex $L_{R/\mathbb{Z}}$, thus we would expect an “absolute” version of cotangent complex, which vanishes on perfect fields of char p . Kanau SHIMADA showed in [Shi26] that, the animation of Saito’s Frobenius–Witt differentials, *Frobenius–Witt cotangent complex*, serves as a good candidate. More precisely, Frobenius–Witt cotangent complex vanishes on perfectoid rings.

The current article is devoted to a direct description of Frobenius–Witt cotangent complex, which leads to generalizations to (nonconnective) derived rings and animated pre-log rings. The starting point is a pullback description of arithmetic extension in [Shi26].

Theorem 1.1. (Definition 2.9 and Remarks 2.10 and 2.11) *Let R be a commutative ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$, M an $R/p = \pi_0(\bar{R})$ -module, and $m \in M$ an element, viewed as a map $\bar{R} \rightarrow M$ of $\bar{R}$ -modules. Then the arithmetic extension $W_2(R, M, m)$ fits into a pullback diagram*

$$\begin{array}{ccc} W_2(R, M, m) & \longrightarrow & R \\ \downarrow & & \downarrow \\ \bar{R} \oplus \mathrm{cofib}(m) & \longrightarrow & \bar{R} \oplus \bar{R}[1] \end{array}$$

of derived rings, where the right vertical map is the composite map

$$R \xrightarrow{\mathrm{id}_R \otimes 1 \otimes 1} R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\mathrm{Frob}} R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = \bar{R} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1]$$

of animated rings.

This description extends directly to derived rings and animated pre-log rings (Proposition 4.6). By virtue of this description, we get a direct construction of Frobenius–Witt cotangent complex.

Theorem 1.2. (Theorems 3.6 and 3.15 and Remark 3.16) *Let R be a animated ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. The commutative diagram*

$$\begin{array}{ccc} \mathbb{Z} & \longrightarrow & R \\ \downarrow & & \downarrow \\ R & \xrightarrow{\psi_R} & \bar{R} \end{array}$$

of derived rings, where ψ_R is the composite map $R \xrightarrow{\text{can}} \bar{R} \xrightarrow{\text{Frob}} \bar{R}$, and the right vertical map $R \rightarrow \bar{R}$ is the canonical map, induces a map

$$\partial_R: L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow L_{\bar{R}/R} = \bar{R}[1]$$

of $\bar{R}$ -modules, which gives rise to a fiber sequence

$$\bar{R} \xrightarrow{m_R} \text{fib}(\partial_R) \longrightarrow L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R}.$$

Then the Frobenius–Witt cotangent complex of R is given by the pointed $\bar{R}$ -module $(\text{fib}(\partial_R), m_R)$.

Again, this description adapts to derived rings and animated pre-log rings (Theorems 4.8 and 4.9). As a consequence of this description, we get a generalization of [Shi26, Prop 6.1] from perfect δ -rings to derived δ -rings.

Theorem 1.3. (Corollary 3.10 and Remark 3.17) *Let R be a derived δ -ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then its Frobenius–Witt cotangent complex is $\bar{R} \oplus L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R}$.*

Moreover, inspired by this description and [AKN23], we introduce Frobenius–Witt cotangent complex relative to derived δ -rings, which coincides with the absolute one when the base is a perfect δ -ring (Lemma 3.25). We thus get a vanishing of relative Frobenius–Witt cotangent complex for (derived) prisms, generalizing the vanishing of Frobenius–Witt cotangent complex of perfectoid rings.

Theorem 1.4. (Theorem 3.26) *Let (A, I) be a (derived) prism, and $R := A/I$ with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then the Frobenius–Witt cotangent complex $FL_{R/A}$ relative to A vanishes.*

Independently of previous developments, we give a regularity criterion via Frobenius–Witt cotangent complex. Unlike relevant results in [Sai22, HJ24, Tak26, Shi26], we assume neither F -finiteness nor locality.

Theorem 1.5. (Theorem 5.2) *Let R be a p -local Noetherian ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then the ring R is regular if and only if its Frobenius–Witt cotangent complex FL_R , as an $\bar{R}$ -module, is flat.*

Warning 1.6. As in [Shi26], Frobenius–Witt cotangent complex is more comparable to Frobenius–Witt differentials “modulo p ” (cf. [Tak26, Def 3.5]) instead of them on the nose, but if we work with derived $\mathbb{Z}_{(p)}$ -algebras, then there is no difference by [Sai22, Lem 1.2(3)].

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2 Arithmetic extension as a pullback

In this section, we give a pull-back description of the arithmetic extension in [Shi26, Def 2.4] (Remark 2.10), which allows us to extend it to derived rings with pointed modules. Technically, we first establish the same result for commutative $\mathbb{F}_p$ -algebras (Proposition 2.6), and then extend to arbitrary commutative rings.

Recall that the *arithmetic extension* $W_2(A, M, m)$ of A by (M, m) , for a commutative ring A , an A/p -module M (concentrated in degree 0), and an element $m \in M$, is defined in [Shi26, Def 2.4] to be a ring, with underlying set $A \times M$, and the addition and multiplication is given by

$$\begin{aligned} (a, x) + (b, y) &= \left(a + b, x + y - \sum_{j=1}^{p-1} \frac{1}{p} \binom{p}{j} a^{p-j} b^j m \right), \\ (a, x) (b, y) &= (ab, a^p y + b^p x), \end{aligned}$$

with multiplicative identity given by $(1, 0)$. We note that $(a, 0) + (0, x) = (a, x)$. We start with the special case that A is a commutative $\mathbb{F}_p$ -algebra.

Example 2.1. Let A be a commutative $\mathbb{F}_p$ -algebra. Then by definition, the ring $W_2(A, A, 1)$ coincides with the ring $W_2(A)$ of 2-truncated (p -typical) Witt vectors. In this case, by [BS22, Rem 2.5], we know that $W_2(A)$ fits into a Cartesian diagram

$$\begin{array}{ccc} W_2(A) & \longrightarrow & A \\ \downarrow & & \downarrow \\ A & \xrightarrow{\text{can}} & A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \end{array}$$

of derived rings, where the right vertical map is the composite map

$$\psi_A : A \xrightarrow{\text{can}} A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\varphi_A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p} A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p.$$

Note that, for any $\mathbb{F}_p$ -algebra A , its derived modulo p admits a structure of trivial square zero extension.

Construction 2.2. Let A be a derived $\mathbb{F}_p$ -algebra. Then the derived ring $A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ can be canonically identified with the square-zero extension $A \oplus A[1]$. Indeed, as

$$A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = A \otimes_{\mathbb{F}_p, 1 \otimes \text{id}_{\mathbb{F}_p}}^{\mathbb{L}} (\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p),$$

it suffices to identify the animated $\mathbb{F}_p$ -algebra $\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$, where the $\mathbb{F}_p$ -algebra structure is given by the second factor, as a square-zero extension $\mathbb{F}_p \oplus \mathbb{F}_p[1]$. We identify the first factor $\mathbb{F}_p$ as the pushout $\mathbb{Z} \otimes_{0 \leftarrow t, \mathbb{Z}[t], t \mapsto p}^{\mathbb{L}} \mathbb{Z}$ of animated rings, and thus the animated $\mathbb{F}_p$ -algebra $\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ can be identified with $\mathbb{F}_p \otimes_{\mathbb{F}_p[t]}^{\mathbb{L}} \mathbb{F}_p = \text{LSym}_{\mathbb{F}_p}(\mathbb{F}_p[1])$, which gives rise to a square-zero extension structure.

Example 2.1 hints the following definition.

Definition 2.3. Let A be a derived $\mathbb{F}_p$ -algebra, and M an A -module spectrum with a map $m : A \rightarrow M$ of A -modules. The map m gives rise to a map $\text{cofib}(m) \rightarrow A[1]$ of A -modules. Then we define $\bar{W}_2(A, M, m)$ to be the pullback of the diagram

$$\begin{array}{ccc} & & A \\ & & \downarrow \psi_A \\ A \oplus \text{cofib}(m) & \longrightarrow & A \oplus A[1] \simeq A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \end{array}$$

of derived rings. As the map ψ_A is not a map of derived $\mathbb{F}_p$ -algebras in general, this diagram is **not** a diagram of derived $\mathbb{F}_p$ -algebras. We denote by res the canonical map $\bar{W}_2(A, M, m) \rightarrow A$.

Remark 2.4. Let A be a derived $\mathbb{F}_p$ -algebra, and M an A -module spectrum with a map $m : A \rightarrow M$ of A -modules. By definition, we have a Cartesian diagram

$$\begin{array}{ccc} \bar{W}_2(A, M, m) & \longrightarrow & A \\ \downarrow & & \downarrow \\ A \oplus \text{cofib}(m) & \longrightarrow & A \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \end{array}$$

of $\mathbb{Z}$ -module spectra, which induces an equivalence of horizontal fibers, and thus we get a fiber sequence

$$M \longrightarrow \bar{W}_2(A, M, m) \xrightarrow{\text{res}} A$$

in $D(\mathbb{Z})$, which is functorial in (A, M, m) . We denote by V the first map $M \rightarrow \bar{W}_2(A, M, m)$. In particular, if both A and M are concentrated in degree 0, then so is the $\mathbb{Z}$ -module spectrum $\bar{W}_2(A, M, m)$, which implies that the derived ring $\bar{W}_2(A, M, m)$ is a classical commutative ring.

Construction 2.5. Let A be a derived $\mathbb{F}_p$ -algebra, and M an A -module spectrum with a map $m : A \rightarrow M$ of A -modules. Apply Remark 2.4 to the map $(A, A, 1) \rightarrow (A, M, m)$ induced by the map m , we get a map

$$\begin{array}{ccccc} A & \xrightarrow{V} & W_2(A) & \xrightarrow{\text{res}} & A \\ \downarrow & & \downarrow & & \parallel \\ M & \xrightarrow{V} & \bar{W}_2(A, M, m) & \xrightarrow{\text{res}} & A \end{array} \quad (2.1)$$

of fiber sequences in $D(\mathbb{Z})$. Since the rightmost vertical map is an equivalence, the left square is Cartesian in $D(\mathbb{Z})$. When both A and M are connective, we get multiplicative inverse $\Omega^\infty A \rightarrow \Omega^\infty \bar{W}_2(A, M, m)$ of the canonical map $\text{res} : \bar{W}_2(A, M, m) \rightarrow A$ given by the composite map

$$[-]_{(A, M, m)} : \Omega^\infty A \xrightarrow{[-]_A} \Omega^\infty W_2(A) \longrightarrow \Omega^\infty \bar{W}_2(A, M, m),$$

where the first map $[-]_A$ is the Teichmüller map, being the multiplicative inverse of the restriction map $\text{res} : W_2(A) \rightarrow A$. Consequently, we get an equivalence

$$\Omega^\infty A \times \Omega^\infty M \longrightarrow \Omega^\infty \bar{W}_2(A, M, m)$$

of anima, informally written as $(a, m) \mapsto ([a]_{(A, M, m)} + V(m))$.

Proposition 2.6. Let A be a commutative $\mathbb{F}_p$ -algebra, and M an A -module concentrated in degree 0, and $m \in M$ an element inducing a map $m : A \rightarrow M$ of A -modules. Then the isomorphism

$$W_2(A, M, m) = A \times M \rightarrow \bar{W}_2(A, M, m)$$

of sets (we invoke Remark 2.4 to see that they are sets, so we omit notations Ω^∞) is compatible with the ring structures on both sides.

Proof. As $(1, 0)$ is the unit of $W_2(A)$, this map preserves multiplicative identities. The additivity is verified by the left Cartesian square of (2.1) in $D(\mathbb{Z})$ in Construction 2.5.

Now we verify the multiplicativity. Note that the map $A \oplus \text{cofib}(m) \rightarrow A \oplus A[1]$ of derived rings in Definition 2.3 can be realized as a square-zero extension, thus so can the (surjective) map $\text{res} : \bar{W}_2(A, M, m) \rightarrow A$ of rings. Given the additivity, the multiplicativity is completely determined by the A -module structure on M , which is given by forgetting the $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ -module structure on M along the map ψ_A . Now M is concentrated in degree 0, so it is completely the $\pi_0(A \otimes_{\mathbb{Z}} \mathbb{F}_p) = A$ -module structure on M , and the composite $A \xrightarrow{\psi_A} A \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\tau \leq 0} \pi_0(A \otimes_{\mathbb{Z}} \mathbb{F}_p) = A$ is simply the Frobenius map φ_A . The multiplicativity then follows. $\square$

Now we extend to the case that A is a general derived ring (in place of a derived $\mathbb{F}_p$ -algebra).

Remark 2.7. Let A be a commutative ring, and M an A/p -module concentrated in degree 0, and $m \in M$ an element. Then we have a map

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & W_2(A, M, m) & \longrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M & \longrightarrow & W_2(A/p, M, m) & \longrightarrow & A/p \longrightarrow 0 \end{array}$$

of short exact sequences, where the rightmost vertical map is the canonical map $A \rightarrow A/p$. It follows that the right square is Cartesian in $D(\mathbb{Z})$ (not just in abelian groups).

We need the following simple observation to see that the difference between $\pi_0(A \otimes_{\mathbb{Z}} \mathbb{F}_p) = A/p$ is $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ is insignificant.

Remark 2.8. Let $f : A \rightarrow B$ be a map of derived $\mathbb{F}_p$ -algebras, and M a B -module, and $m : B \rightarrow M$ a map of B -modules. Then by Remark 2.4, the square

$$\begin{array}{ccc} \bar{W}_2(A, M, m) & \longrightarrow & A \\ \downarrow & & \downarrow \\ \bar{W}_2(B, M, m \circ f) & \longrightarrow & B \end{array}$$

of derived rings is Cartesian.

We come to the following definition.

Definition 2.9. Let A be a derived ring, and M an $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ -module, and $m: A \otimes_{\mathbb{Z}} \mathbb{F}_p \rightarrow M$ a map of $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ -modules. Then we define the derived arithmetic extension $W_2^{\text{der}}(A, M, m)$ (which will be denoted by $W_2(A, M, m)$ after Remark 2.11) of A by M to be the pullback of the diagram

$$\begin{array}{ccc} & & A \\ & & \downarrow \\ \overline{W}_2(A \otimes_{\mathbb{Z}} \mathbb{F}_p, M, m) & \longrightarrow & A \otimes_{\mathbb{Z}} \mathbb{F}_p \end{array}$$

of derived rings, where the right vertical map is the canonical map $\text{can}: A \rightarrow A \otimes_{\mathbb{Z}} \mathbb{F}_p$ (not the Frobenius). We denote by res the canonical map $W_2^{\text{der}}(A, M, m) \rightarrow A$.

Remark 2.10. Let A be a derived ring with $\overline{A} := A \otimes_{\mathbb{Z}} \mathbb{F}_p$, and M an $\overline{A}$ -module, and $m: \overline{A} \rightarrow M$ a map of $\overline{A}$ -modules. Unraveling definitions, we see that the derived ring $W_2^{\text{der}}(A, M, m)$ fits canonically into a Cartesian diagram

$$\begin{array}{ccc} W_2^{\text{der}}(A, M, m) & \longrightarrow & A \\ \downarrow & & \downarrow \\ \overline{A} \oplus \text{cofib}(m) & \longrightarrow & \overline{A} \oplus \overline{A}[1] \end{array},$$

where the right vertical map is the composite $A \xrightarrow{\text{can}} \overline{A} \xrightarrow{\psi_{\overline{A}}} \overline{A} \otimes_{\mathbb{Z}} \mathbb{F}_p = \overline{A} \oplus \overline{A}[1]$, and the bottom horizontal map is induced by the boundary map $\text{cofib}(m) \rightarrow \overline{A}[1]$.

Remark 2.11. Let A be a commutative ring, and M an A/p -module concentrated in degree 0, and $m \in M$ an element. Remark 2.8 shows that the square

$$\begin{array}{ccc} \overline{W}_2(A \otimes_{\mathbb{Z}} \mathbb{F}_p, M, m) & \longrightarrow & A \otimes_{\mathbb{Z}} \mathbb{F}_p \\ \downarrow & & \downarrow \\ \overline{W}_2(A/p, M, m) & \longrightarrow & A/p \end{array}$$

of derived rings is Cartesian. Thus by Remark 2.7, it follows from Proposition 2.6 that the functor W_2^{der} defined in Definition 2.9 coincides with the functor W_2 in [Shi26, Def 2.4] reviewed above when derived rings and modules in question are concentrated in degree 0, thus we will denote W_2^{der} by W_2 in the sequel.

Remark 2.12. Let A be a derived $\mathbb{F}_p$ -algebra, M an A -module, and $m: A \rightarrow M$ a map of A -modules. The multiplication map $\mu_A: A \otimes_{\mathbb{Z}} \mathbb{F}_p \rightarrow A$ endows M an $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ -module structure. Now we examine the diagram

$$\begin{array}{ccc} W_2(A, M, m) & \longrightarrow & A \\ \downarrow & & \downarrow \\ \overline{W}_2(A \otimes_{\mathbb{Z}} \mathbb{F}_p, M, m \circ \mu_A) & \longrightarrow & A \otimes_{\mathbb{Z}} \mathbb{F}_p \\ \downarrow & & \downarrow \\ \overline{W}_2(A, M, m) & \longrightarrow & A \end{array}$$

of derived rings. By definition, the top square is Cartesian. By Remark 2.8, the bottom square is Cartesian as well, thus so is the outer square. Since the composite map

$$A \xrightarrow{\text{can}} A \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\mu_A} A$$

is an equivalence, it follows that the map $W_2(A, M, m) \rightarrow \overline{W}_2(A, M, m)$ is an equivalence as well.

Remark 2.13. Let A be a derived ring, and M an $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ -module, and $m: A \otimes_{\mathbb{Z}} \mathbb{F}_p \rightarrow M$ a map of $A \otimes_{\mathbb{Z}} \mathbb{F}_p$ -modules. By Definition 2.9 and Remark 2.4, we have a fiber sequence

$$M \longrightarrow W_2(A, M, m) \xrightarrow{\text{res}} A$$

in $D(\mathbb{Z})$. We denote by V the first map $M \rightarrow W_2(A, M, m)$. This shows that the functor W_2 preserves sifted colimits, and in particular, if both A and M are connective, then so is $W_2(A, M, m)$, and this coincides canonically with $W_2^{\text{an}}(A, M, m)$ in [Shi26, Def 4.1].

Remark 2.14. Let $f: A \rightarrow B$ be a map of derived rings, and M a B -module, and $m: B \rightarrow M$ a map of B -modules. Then by Remark 2.4, the square

$$\begin{array}{ccc} W_2(A, M, m) & \longrightarrow & A \\ \downarrow & & \downarrow \\ W_2(B, M, m \circ f) & \longrightarrow & B \end{array}$$

of derived rings is Cartesian.

3 Frobenius–Witt cotangent complex relative to δ -rings

In this section, thanks to the pullback description in Remark 2.10, we give a rather direct description of Frobenius–Witt cotangent complex (Theorems 3.6 and 3.15), which extends directly to derived rings. As a consequence, we deduce a computation of Frobenius–Witt cotangent complex of derived δ -rings (Corollary 3.10 and Remark 3.17), generalizing the result [Shi26, Prop 6.1] for perfect δ -rings. Inspired by this description, we also introduce the concept of Frobenius–Witt cotangent complex relative to derived δ -rings (Definition 3.19), and deduce a generalization of vanishing of Frobenius–Witt cotangent complex of perfectoid rings (Theorem 3.26).

Construction 3.1. Let R be a derived $\mathbb{F}_p$ -algebra. The map $\psi_R: R \rightarrow R \otimes_{\mathbb{Z}} \mathbb{F}_p = R \oplus R[1]$ of derived rings (here we are invoking Construction 2.2) is then classified by a map $\bar{\partial}_R: L_{R/\mathbb{Z}} \otimes_{R, \varphi_R}^{\mathbb{L}} R \rightarrow R[1]$ of R -modules, where we use the fact that the composite map

$$R \xrightarrow{\psi_R} R \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\mu_R} R$$

coincides with the Frobenius map φ_R . This gives rise to a map $\bar{m}_R: R \rightarrow \text{fib}(\bar{\partial}_R)$ of R -modules, along with a fiber sequence

$$R \xrightarrow{\bar{m}_R} \text{fib}(\bar{\partial}_R) \longrightarrow L_{R/\mathbb{Z}} \otimes_{R, \varphi_R}^{\mathbb{L}} R$$

of R -modules.

Lemma 3.2. *The map $\bar{\partial}_{\mathbb{F}_p}$ of $\mathbb{F}_p$ -modules in Construction 3.1 is an equivalence.*

Proof. Note that the source and the target of $\bar{\partial}_{\mathbb{F}_p}$ are both $\mathbb{F}_p[1]$, thus $\bar{\partial}_{\mathbb{F}_p}$ is either 0 or an equivalence. It suffices to exclude the possibility of $\bar{\partial}_{\mathbb{F}_p} = 0$. Indeed, if it were zero, then the diagram $\mathbb{F}_p \xrightarrow{\text{can}} \mathbb{F}_p \oplus \mathbb{F}_p[1] \xleftarrow{\psi_{\mathbb{F}_p}} \mathbb{F}_p$ would be the trivial square-zero extension $\mathbb{F}_p \oplus \mathbb{F}_p$, while it is actually $W_2(\mathbb{F}_p) = \mathbb{Z}/p^2$. $\square$

Construction 3.3. Let R be a derived ring. Apply Construction 3.1 to the derived $\mathbb{F}_p$ -algebra $\bar{R} = R \otimes_{\mathbb{Z}} \mathbb{F}_p$, we get a composite map

$$\partial_R: L_{R/\mathbb{Z}} \otimes_{R, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow L_{\bar{R}/\mathbb{Z}} \otimes_{\bar{R}, \varphi_{\bar{R}}}^{\mathbb{L}} \bar{R} \xrightarrow{\bar{\partial}_{\bar{R}}} \bar{R}[1]$$

of $\bar{R}$ -modules classifying the composite map $R \xrightarrow{\text{can}} \bar{R} \xrightarrow{\psi_R} \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1]$ of derived rings, which gives rise to a map $m_R: \bar{R} \rightarrow \text{fib}(\partial_R)$ of $\bar{R}$ -modules, along with a fiber sequence

$$\bar{R} \xrightarrow{m_R} \text{fib}(\partial_R) \longrightarrow L_{R/\mathbb{Z}} \otimes_{R, \psi_R}^{\mathbb{L}} \bar{R}$$

of $\bar{R}$ -modules.

Lemma 3.4. *The map $\partial_{\mathbb{F}_p}$ in Construction 3.3 is an equivalence.*

Proof. Let $R := \mathbb{F}_p$, and $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. The map $\bar{R} \rightarrow R$ of derived $\mathbb{F}_p$ -algebras gives rise to a commutative diagram

$$\begin{array}{ccccc} R & \xrightarrow{\text{can}} & \bar{R} & \xrightarrow{\psi_{\bar{R}}} & \bar{R} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1] \\ & & \downarrow & & \downarrow \\ & & R & \xrightarrow{\psi_R} & R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = R \oplus R[1] \end{array}$$

of derived rings (the commutativity of the rightmost square uses the map $\bar{R} \rightarrow R$ of derived $\mathbb{F}_p$ -algebras, not merely as a map of derived rings). It follows that $\bar{\partial}_R = \partial_R \otimes_{\bar{R}}^{\mathbb{L}} R$. The result then follows from Lemma 3.2 the conservativity of the functor $(-) \otimes_{\bar{R}}^{\mathbb{L}} R : D(\bar{R})^{\omega} \rightarrow D(R)^{\omega}$ on perfect complexes. $\square$

As in [Shi26, Def 4.5], we define Frobenius–Witt derivations in the derived context.

Definition 3.5. *Let R be a derived ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$, and M an $\bar{R}$ -module, and $m : \bar{R} \rightarrow M$ a map of $\bar{R}$ -modules. Then the anima $\text{FWDer}(R, (M, m))$ of Frobenius–Witt derivations is defined to be*

$$\text{Hom}_{\text{DAI}_{\mathbb{G}/R}}(R, W_2(R, M, m)),$$

where the map $W_2(R, M, m) \rightarrow R$ is the canonical map res in Definition 2.9.

The following statement shows that $(\text{fib}(\partial_R), m_R)$ coincides with Frobenius–Witt cotangent complex in [Shi26, Prop 4.8] for animated $\mathbb{Z}_{(p)}$ -algebras R , and in particular, it could be viewed as the definition of Frobenius–Witt cotangent complex for derived $\mathbb{Z}_{(p)}$ -algebras. In particular, this gives rise to the vanishing of Frobenius–Witt cotangent complex [Shi26, Lem 6.3] without computations.

Theorem 3.6. *Let R be a derived ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then the functor*

$$\begin{aligned} D(\bar{R})_{\bar{R}/} &\longrightarrow \text{An} \\ (M, m) &\longmapsto \text{FWDer}(R, (M, m)) \end{aligned}$$

is co-represented by $(\text{fib}(\partial_R), m_R)$ in Construction 3.3.

Proof. Let M be an $\bar{R}$ -module, and $m : \bar{R} \rightarrow M$ a map of $\bar{R}$ -modules. Unraveling definitions, by Remark 2.10, we see that the anima $\text{FWDer}(R, (M, m))$ is given by the anima of a map

$$\alpha : L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow \text{cofib}(m)$$

along with a homotopy between the map $\partial_R : L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R} \rightarrow \bar{R}[1]$ of $\bar{R}$ -modules in Construction 3.3 and the composite map

$$L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R} \xrightarrow{\alpha} \text{cofib}(m) \longrightarrow \bar{R}[1]$$

of $\bar{R}$ -modules, i.e. the anima of factorizations of ∂_R through $\text{cofib}(m) \rightarrow \bar{R}[1]$ in $D(\bar{R})$. On the other hand, as the ∞ -category $D(\bar{R})$ is stable, we have an equivalence

$$\begin{aligned} D(\bar{R})_{\bar{R}/} &\longrightarrow D(\bar{R})_{\bar{R}[1]} \\ (\beta : \bar{R} \rightarrow M) &\longmapsto (\text{cofib}(\beta) \rightarrow \bar{R}[1]) \end{aligned}$$

of ∞ -categories, and in particular,

$$\text{Hom}_{D(\bar{R})_{\bar{R}/}}((\text{fib}(\partial_R), m_R), (M, m)) = \text{Hom}_{D(\bar{R})_{\bar{R}[1]}}(\partial_R, \text{cofib}(m) \rightarrow \bar{R}[1])$$

where the map $\text{cofib}(m) \rightarrow \bar{R}[1]$ is the boundary map. The result then follows. $\square$

Such a description would be helpful even if we restrict to animated rings or even classical rings, as long as we have a way to control its Frobenius. For example, the Frobenius map of δ -rings are well behaved. We have the following simple observation.

Lemma 3.7. *Let R be a derived δ -ring, and S a derived $\mathbb{F}_p$ -algebra. Then the Frobenius map $\varphi_{R \otimes_{\mathbb{Z}} S} : R \otimes_{\mathbb{Z}} S \rightarrow R \otimes_{\mathbb{Z}} S$ can be canonically identified with the map $\varphi_R \otimes_{\mathbb{Z}} \varphi_S$.*

Proof. Let $\bar{R}$ be the derived $\mathbb{F}_p$ -algebra $R \otimes_{\mathbb{Z}} \mathbb{F}_p$. Then

$$\begin{aligned} \varphi_R \otimes_{\mathbb{Z}} \varphi_S &= \varphi_{\bar{R} \otimes_{\mathbb{F}_p} S} \\ &= \varphi_{\bar{R} \otimes_{\mathbb{F}_p} S} \\ &= \varphi_{R \otimes_{\mathbb{Z}} S}, \end{aligned}$$

where the first equality follows from the fact that φ_R is a derived Frobenius lift. $\square$

Corollary 3.8. *Let R be a derived δ -ring. Then the map ∂_R is canonically null-homotopic.*

Proof. By Lemma 3.7 (with $S = \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p$), we see that the composite map

$$R \xrightarrow{\text{can}} \bar{R} \xrightarrow{\psi_R} \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1]$$

of derived rings can be rewritten as the composite of the map

$$\alpha : R \xrightarrow{\text{id}_R \otimes 1 \otimes 1} R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\text{id}_R \otimes \varphi_{\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}} \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1]$$

of derived rings with the map

$$\varphi_{\bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p} : \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{(\varphi_R \otimes_{\mathbb{Z}} \mathbb{F}_p) \otimes_{\mathbb{Z}} \mathbb{F}_p} \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p$$

of derived $\mathbb{F}_p$ -algebras. Note that the map $\beta : L_{R/\mathbb{Z}} \otimes_{R, \text{can}} \bar{R} \rightarrow \bar{R}[1]$ classifying the map α into the trivial square-zero extension $\bar{R} \oplus \bar{R}[1]$ is canonically null-homotopic, as the initiality of $\mathbb{Z}$ in DA_{lg} implies that any map $\mathbb{Z} \rightarrow \mathbb{F}_p \otimes \mathbb{F}_p = \mathbb{F}_p \oplus \mathbb{F}_p[1]$ of derived rings coincides with the composite map $\mathbb{Z} \rightarrow \mathbb{F}_p \hookrightarrow \mathbb{F}_p \oplus \mathbb{F}_p[1]$ of derived rings. It follows that $\partial_R = \beta \otimes_{R, \varphi_R} \bar{R}$ is canonically null-homotopic as well. $\square$

Warning 3.9. Let R be a derived ring. Then the composite map

$$R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\text{id}_R \otimes \varphi_{\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}} R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\varphi_{R \otimes_{\mathbb{Z}} \mathbb{F}_p} \otimes \text{id}_{\mathbb{F}_p}} R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p$$

of derived $\mathbb{F}_p$ -algebras is well-defined. In the proof of Corollary 3.8, we show that, a δ -structure on R gives rise to an identification of this composite map with the Frobenius map $\varphi_{R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}$ via Lemma 3.7. However, there is no such identification $R = \mathbb{F}_p$, as if there were, then the proof of Lemma 3.7 would imply that $\partial_R = 0$, but we have already seen in Lemma 3.4 that $\partial_{\mathbb{F}_p}$ is an automorphism of a non-zero spectrum.

It follows from Corollary 3.8 that

Corollary 3.10. *Let R be a derived δ -ring, and $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$. Then the Frobenius–Witt cotangent complex of R is $\bar{R} \oplus L_{R/\mathbb{Z}} \otimes_{R, \psi_R} \bar{R}$, with m being the inclusion $\bar{R} \hookrightarrow \bar{R} \oplus L_{R/\mathbb{Z}} \otimes_{R, \psi_R} \bar{R}$.*

Example 3.11. Let R be a perfect δ -ring. Then $L_{R/\mathbb{Z}} \otimes_{R, \text{can}} \bar{R} = L_{\bar{R}/\mathbb{F}_p} = 0$. By Corollary 3.10 (and the fact that $\psi_R = \varphi_{\bar{R}} \circ \text{can}$), the Frobenius–Witt cotangent complex of R is $\bar{R}$ with $m = 1$, recovering [Shi26, Prop 6.1]. Thus Corollary 3.10 is a generalization of it to non-perfect (derived) δ -rings.

We now give a simple description of the Frobenius map $\varphi_{\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}$, which simplifies the map ∂_R in Construction 3.3 a lot. We first explain how to get a filtered Frobenius map on filtered derived $\mathbb{F}_p$ -algebras. One may compare this construction with [AMMN22, Ex A.11].

Construction 3.12. (Filtered Frobenius) We construct filtered Frobenius map on $(\mathbb{N}, \geq)$ -filtered derived $\mathbb{F}_p$ -algebras as in [Hol23, Cons 2.4.1]. Indeed, given an $(\mathbb{N}, \geq)$ -filtered commutative $\mathbb{F}_p$ -algebra $\text{Fil } A$, we simply assign $x \in \text{Fil}^m A$ to $x^p \in \text{Fil}^{mp} A$. This gives rise to a map

$$\varphi_{\text{Fil } A} : \text{Fil } A \longrightarrow (\text{Fil}^{mp} A)_{m \in \mathbb{N}} \quad (3.1)$$

of $(\mathbb{N}, \geq)$ -filtered commutative $\mathbb{F}_p$ -algebras. We restrict such construction to free filtered commutative $\mathbb{F}_p$ -algebras generated on a finitely filtered finite set, and argue as in [Hol23, Cons 2.4.1], we see that this construction is right-extendable. This extends the filtered Frobenius map $\varphi_{\text{Fil} A}$ in (3.1) to $(\mathbb{N}, \geq)$ -filtered derived $\mathbb{F}_p$ -algebras $\text{Fil} A$.

Now we observe that, the Frobenius map on a square-zero extension factors through the extended algebra.

Construction 3.13. Let R be a derived $\mathbb{F}_p$ -algebra, M an R -module, and S a square-zero extension of R by M within $D(\mathbb{F}_p)$. By [Mag24, Prop 3.3.17], we may identify S with an $(\mathbb{N}, \geq)$ -filtered derived $\mathbb{F}_p$ -algebra $\text{Fil} S$ concentrated in weights $[0, 1]$, such that $\text{Fil}^0 S = S$, $\text{Fil}^1 S = M$, and $\text{gr}^0 S = R$. We now apply Construction 3.12 to the filtered derived $\mathbb{F}_p$ -algebra $\text{Fil} S$, obtaining the filtered Frobenius map $\varphi_{\text{Fil} S}$. Note that the filtered derived $\mathbb{F}_p$ -algebra $(\text{Fil}^{mp} S)_{m \in \mathbb{N}}$ is concentrated in weights $\{0\}$. It follows that the Frobenius map $\varphi_S: S \rightarrow S$ coincides with the composite

$$\text{Fil}^0 S \longrightarrow \text{gr}^0 S \xrightarrow{\varphi_{\text{Fil} S}} \text{gr}^0 (\text{Fil}^{mp} S)_{m \in \mathbb{N}} \xleftarrow{\simeq} \text{Fil}^0 S.$$

This gives rise to a factorization $S \rightarrow R \rightarrow S$ of the Frobenius map φ_S . In particular, we get a factorization

$$\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\text{mult}} \mathbb{F}_p \xrightarrow{1 \otimes \text{id}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$$

of the Frobenius map $\varphi_{\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}$, where the $\mathbb{F}_p$ -algebra structure on each tensor product is given by right factor.

Remark 3.14. Construction 3.13 gives rise to another proof of Lemma 3.2. Indeed, the map $\bar{\partial}_{\mathbb{F}_p}$ classifies the map $\mathbb{F}_p \xrightarrow{1 \otimes \text{id}_{\mathbb{F}_p}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = \mathbb{F}_p \oplus \mathbb{F}_p[1]$ (cf. Construction 2.2). This is the base change of the map

$$\mathbb{Z} \xrightarrow{1 \otimes \text{id}_{\mathbb{Z}}} \mathbb{Z} \otimes_{\mathbb{Z}[t]}^{\mathbb{L}} \mathbb{Z} = \mathbb{Z} \oplus \mathbb{Z}[1]$$

of animated $\mathbb{Z}[t]$ -algebras along $\mathbb{Z}[t] \rightarrow \mathbb{Z}, t \mapsto p$, and thus it suffices to see that the classifying map $\gamma: L_{\mathbb{Z}/\mathbb{Z}[t]} \rightarrow \mathbb{Z}[1]$ is an equivalence. The coCartesian square

$$\begin{array}{ccc} \mathbb{Z}[t] & \longrightarrow & \mathbb{Z} \\ \downarrow & & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathbb{Z} \otimes_{\mathbb{Z}[t]}^{\mathbb{L}} \mathbb{Z} \end{array}$$

in $\text{DA}_{\mathbb{Z}}$ is LSym of the (co)Cartesian square

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\alpha} & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z}[1] \end{array}$$

in $D(\mathbb{Z})$, and thus the map $L_{\mathbb{Z}/\mathbb{Z}[t]} \rightarrow \mathbb{Z}[1]$ is actually induced by the map $\beta: \text{cofib}(\alpha) \rightarrow \mathbb{Z}[1]$, which is subsequently induced by the top-right part of the previous Cartesian square. Since the map β is an equivalence, it follows that the map γ is an equivalence as well.

Construction 3.13 allows us to simplify the map ∂_R in Construction 3.3.

Theorem 3.15. *Let R be a derived ring. Then the map ∂_R in Construction 3.3 can be identified with the composite map*

$$L_{R/\mathbb{Z}} \otimes_{R, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow L_{\bar{R}/\mathbb{Z}} \longrightarrow L_{\bar{R}/R} \longrightarrow \bar{R}[1] \quad (3.2)$$

in $D(\bar{R})$, where the first map is part of the transitivity sequence of $\mathbb{Z} \rightarrow R \xrightarrow{\psi_R} \bar{R}$, and the last map is classified by the map

$$\bar{R} = R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Z} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\text{id}_R \otimes 1 \otimes \text{id}_{\mathbb{F}_p}} R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = \bar{R} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1]$$

of derived rings. Moreover, the last map $L_{\bar{R}/R} \rightarrow \bar{R}[1]$ in (3.2) is an equivalence.

Proof. First, it follows from Remark 3.14 that the last claim $L_{\bar{R}/R} \rightarrow \bar{R}[1]$ in (3.2). Let R be a derived ring. Now we analyze the composite map

$$R \xrightarrow{\text{id}_R \otimes 1 \otimes 1} R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\varphi_{R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}} R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p \quad (3.3)$$

classified by the map ∂_R in Construction 3.3. The derived $\mathbb{F}_p$ -algebra $R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p$ can be rewritten as $\bar{R} \otimes_{\mathbb{F}_p, 1 \otimes \text{id}} (\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p)$, where the $\mathbb{F}_p$ -algebra structure on $\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p$ is given by the second factor, and the tensor product is also over it. Then the Frobenius map $\varphi_{R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}$ is identified with the map $\varphi_{\bar{R}} \otimes \varphi_{\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p}$. Now applying Construction 3.13, we see that the composite map (3.3) can be rewritten as

$$R \xrightarrow{\psi_R} R \otimes_{\mathbb{Z}} \mathbb{F}_p = R \otimes_{\mathbb{Z}} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{F}_p \xrightarrow{\text{id}_R \otimes 1 \otimes \text{id}_{\mathbb{F}_p}} R \otimes_{\mathbb{Z}} \mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{F}_p.$$

The result then follows. $\square$

Remark 3.16. In retrospect, it is conceptually better to interpret the composite of the first two maps in (3.2) as the map induced by the commutative diagram

$$\begin{array}{ccc} \mathbb{Z} & \longrightarrow & R \\ \downarrow & & \downarrow \\ R & \xrightarrow{\psi_R} & \bar{R} \end{array}$$

of derived rings, where the right vertical map is the canonical map $\text{can} : R \rightarrow \bar{R}$, via Construction 3.18, as in the relative case (Remark 3.23).

Remark 3.17. We can reproduce Corollary 3.8 via Theorem 3.15 as follows. Let R be a derived δ -ring. Then the first map in (3.2) coincides with the composite map

$$L_{R/\mathbb{Z}} \otimes_{\bar{R}, \psi_R} \bar{R} \longrightarrow L_{R/\mathbb{Z}} \otimes_{\bar{R}, \text{can}} \bar{R} \longrightarrow L_{\bar{R}/\mathbb{Z}}$$

in $D(\bar{R})$, where the first map is induced by φ_R . Now the result follows from the null-homotopy of the transitivity sequence

$$L_{R/\mathbb{Z}} \otimes_{\bar{R}, \text{can}} \bar{R} \longrightarrow L_{\bar{R}/\mathbb{Z}} \longrightarrow L_{\bar{R}/R}$$

associated to the maps $\mathbb{Z} \rightarrow R \xrightarrow{\text{can}} \bar{R}$ of derived rings.

Inspired by this description of Frobenius–Witt cotangent complex and prismatic cohomology relative to δ -rings introduced in [AKN23], we introduce the following relative version of ∂_R in Construction 3.3.

Construction 3.18. Let A be a derived δ -ring, and R a derived A -algebra. Denote $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$. Then the map $\psi_{\bar{R}} : \bar{R} \rightarrow \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p$ of derived rings fits into a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\varphi_A} & A \\ \downarrow & & \downarrow \\ R & \xrightarrow{\psi_R} & \bar{R} \end{array}$$

of derived rings, and thus the composite map

$$R \xrightarrow{\text{can}} \bar{R} \xrightarrow{\psi_{\bar{R}}} \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p = \bar{R} \oplus \bar{R}[1]$$

of derived rings (here we are invoking Construction 2.2) carries a canonical A -algebra structure (where the A -algebra structure on the target is forgotten along φ_A). Thus it induces a map

$$\partial_{R/A} : L_{R/A} \otimes_{\bar{R}, \psi_R} \bar{R} \longrightarrow \bar{R}[1]$$

in $D(\bar{R})$. We also have a map $m_{R/A} : \bar{R} \rightarrow \text{fib}(\partial_{R/A})$ in $D(\bar{R})$, along with a fiber sequence

$$\bar{R} \xrightarrow{m_{R/A}} \text{fib}(\partial_{R/A}) \longrightarrow L_{R/A} \otimes_{\bar{R}, \psi_R} \bar{R}.$$

This leads to Frobenius–Witt cotangent complex relative to derived δ -rings.

Definition 3.19. Let A be a derived δ -ring, and R a derived A -algebra with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$.

- The Frobenius–Witt cotangent complex $FL_{R/A}$ relative to A is the pointed $\bar{R}$ -module $(\text{fib}(\partial_{R/A}), m_{R/A})$ as in Construction 3.18.
- Let M be an $\bar{R}$ -module, and $m: \bar{R} \rightarrow M$ a map of $\bar{R}$ -modules. Then the anima $\text{FWDer}_A(R, (M, m))$ of Frobenius–Witt derivations relative to A is defined to be

$$\text{Hom}_{\text{DAlg}_{A/\cdot/R}}(R, W_2(R, M, m)),$$

where the map $W_2(R, M, m) \rightarrow R$ is the canonical map res in Definition 2.9, and the A -algebra structure on $W_2(R, M, m)$ is induced by the A -algebra structure on R , and forgetting the A -algebra structure on $\bar{R} \oplus \text{cofib}(m) \rightarrow \bar{R} \oplus \bar{R}[1]$ along $\varphi_A: A \rightarrow A$.

When everything is concentrated in degree 0, we get a down-to-earth description of Frobenius–Witt derivations relative to δ -rings.

Example 3.20. Let A be a δ -ring, and R a commutative A -algebra, and M a A/p -module, and $m \in M$ an element. We see that the map $A \rightarrow W_2(R, M, m)$ of commutative rings is concretely given by $a \mapsto (a, \delta(a)m)$. Thus a Frobenius–Witt derivation $R \rightarrow M$ relative to A can be identified with a Frobenius–Witt derivation $d: R \rightarrow M$ such that $d(a) = \delta(a)d(p)$ for every $a \in A$.

Applying the proof of Theorem 3.6, we see that

Proposition 3.21. Let A be a derived δ -ring, R be a derived A -algebra with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then the functor

$$\begin{aligned} D(\bar{R})_{\bar{R}/} &\longrightarrow \text{An} \\ (M, m) &\longmapsto \text{FWDer}_A(R, (M, m)) \end{aligned}$$

is co-represented by $(\text{fib}(\partial_{R/A}), m_{R/A})$ in Construction 3.18.

Construction 3.22. Let

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ R & \longrightarrow & S \end{array} \tag{3.4}$$

be a commutative diagram of derived rings. Then we get a map

$$c: L_{R/A} \otimes_R^{\mathbb{L}} S \longrightarrow L_{S/B}$$

of S -modules. When the commutative square (3.4) is coCartesian, the map c is an equivalence.

Remark 3.23. Let A be a derived δ -ring, and R a derived A -algebra with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then as in Theorem 3.15, the map $\partial_{R/A}$ can be identified with the composite map

$$L_{R/A} \otimes_{R, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow L_{\bar{R}/A} \longrightarrow L_{\bar{R}/R} \xrightarrow{\simeq} \bar{R}[1],$$

where the first two maps are induced as in Construction 3.22 by the two commutative squares

$$\begin{array}{ccccc} A & \xrightarrow{\varphi_A} & A & \longrightarrow & R \\ \downarrow & & \downarrow & & \downarrow \\ R & \xrightarrow{\psi_R} & \bar{R} & \xrightarrow{\text{id}_{\bar{R}}} & \bar{R} \end{array} \tag{3.5}$$

of derived rings, where the middle and the rightmost vertical maps are canonical maps (no Frobenius involved). In short, the map $\partial_{R/A}$ can be identified with the map in Construction 3.22 associated to the outer square of (3.5).

We give a relative and derived generalization of transitivity sequence of Frobenius–Witt cotangent complex in [Shi26, Thm 5.3].

Construction 3.24. Let A be a derived δ -ring, and $R \rightarrow S$ a map of derived A -algebras, with $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$ and $\bar{S} := S \otimes_{\mathbb{Z}} \mathbb{F}_p$. Then by Remark 3.23 we get a commutative diagram

$$\begin{array}{ccc} L_{R/A} \otimes_{R, \psi_R}^{\mathbb{L}} \bar{R} \otimes_{\bar{R}}^{\mathbb{L}} \bar{S} & \xrightarrow{\partial_{R/A} \otimes_{\bar{R}}^{\mathbb{L}} \bar{S}} & L_{\bar{R}/R} \otimes_{\bar{R}}^{\mathbb{L}} \bar{S} \\ \downarrow & & \downarrow \simeq \\ L_{S/A} \otimes_{S, \psi_S}^{\mathbb{L}} \bar{S} & \xrightarrow{\partial_{S/A}} & L_{\bar{S}/S} \end{array}$$

in $D(\bar{S})$. The “octahedral” axiom gives rise to a fiber sequence

$$FL_{R/A} \otimes_{\bar{R}}^{\mathbb{L}} \bar{S} \longrightarrow FL_{S/A} \longrightarrow L_{S/R} \otimes_{S, \psi_S}^{\mathbb{L}} \bar{S}$$

in $D(\bar{S})$, called the *transitivity sequence* of (relative) Frobenius–Witt cotangent complex.

Lemma 3.25. Let $A \rightarrow B$ be a map of animated δ -rings such that the induced map $A \otimes_{\mathbb{Z}} \mathbb{F}_p \rightarrow B \otimes_{\mathbb{Z}} \mathbb{F}_p$ of animated $\mathbb{F}_p$ -algebras is relatively perfect. Let R be a derived B -algebra with $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$. Then the canonical map $FL_{R/A} \rightarrow FL_{R/B}$ in $D(\bar{R})_{\bar{R}}$ is an equivalence.

In particular, let A be a perfect δ -ring, and R a derived A -algebra. Then the Frobenius–Witt cotangent complex $FL_{R/A}$ relative to A coincides with the absolute Frobenius–Witt cotangent complex FL_R .

Proof. Since the map $A \otimes_{\mathbb{Z}} \mathbb{F}_p \rightarrow B \otimes_{\mathbb{Z}} \mathbb{F}_p$ of animated $\mathbb{F}_p$ -algebras is relatively perfect, the cotangent complex $L_{B/A}$ is p -completely zero ([GR03, Lem 6.5.9(i)], cf. [Mao25, Lem 3.21] or the proof of [BF26, Lem 2.19]), and thus the canonical map $L_{R/A} \rightarrow L_{R/B}$ in $D(R)$ is p -completely an equivalence. The result then follows from Remark 3.23, as $\bar{R}$ is a derived $\mathbb{F}_p$ -algebra. $\square$

[Shi26, Thm 6.8] establishes vanishing of Frobenius–Witt cotangent complex of perfectoid rings. Thanks to the concrete description in Remark 3.23, we have a relative generalization (it is a generalization by Lemma 3.25).

Theorem 3.26. Let (A, I) be a derived prism, and $R := A/I$ with $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$. Then the map

$$\partial_{R/A} : L_{R/A} \otimes_{R, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow \bar{R}[1]$$

of $\bar{R}$ -modules is an equivalence. Consequently, the Frobenius–Witt cotangent complex $FL_{R/A}$ relative to A vanishes.

Proof. By Remark 3.23, it suffices to see that the composite commutative square

$$\begin{array}{ccccc} A & \xrightarrow{\varphi_A} & A & \longrightarrow & R \\ \downarrow & & \downarrow & & \downarrow \\ R & \xrightarrow{\psi_R} & \bar{R} & \xrightarrow{\text{id}} & \bar{R} \end{array}$$

in $\text{DA}lg$ is coCartesian. This is precisely distinguishedness of the derived pre-prism $I \rightarrow A$: the canonical map $A/\mathbb{L}(I, \varphi^*(I)) \rightarrow A/\mathbb{L}(p, I)$ is an equivalence (cf. [Mao24, §2.4], a derived generalization of [BL22, Rem 2.1.5]). $\square$

4 Logarithmic Frobenius–Witt cotangent complex

For a pre-log ring $(R, P_R, (\alpha_R : P_R \rightarrow (R, \cdot)))$ and an R/p -module M , a *logarithmic Frobenius–Witt derivation* of (R, P_R, α_R) valued in M , defined in [Tak26, Def 3.2], is a pair (d, d^b) of maps, where $d : R \rightarrow M$ is a Frobenius–Witt derivation, and $d^b : P_R \rightarrow M$ is a map of abelian monoids satisfying

$$d(\alpha_R(a)) = \alpha_R(a)^p d^b(a)$$

for every $a \in P_R$. Recall that an *animated pre-log ring* (cf. [BLPØ23, §2.6]) is a triple (R, P_R, α_R) where R is an animated ring, P_R is an animated abelian monoid, and $\alpha_R: P_R \rightarrow (\Omega^\infty R, \cdot)$ a map of animated abelian monoids. We denote by $\mathbf{LAlg}^{\text{an}}$ the ∞ -category of animated pre-log rings.

We explain that previous considerations lead to a “concrete” definition of logarithmic Frobenius–Witt cotangent complex of animated pre-log rings as well. First, the arithmetic extension is canonically equipped with a pre-log structure.

Definition 4.1. Let (R, P_R, α_R) be a pre-log ring, and M an R/p -module, and $m \in M$ an element. The arithmetic extension of (R, P_R, α_R) by M is the pre-log ring $(W_2(R, M, m), P_R \oplus M, \beta_R)$, where the map $\beta_R: P_R \oplus M \rightarrow (W_2(R, M, m), \cdot)$ induced by the map

$$\begin{aligned} P_R &\longrightarrow W_2(R, M, m) \\ a &\longmapsto (\alpha_R(a), 0) \end{aligned}$$

and the map

$$\begin{aligned} M &\longrightarrow W_2(R, M, m) \\ x &\longmapsto (1, x) \end{aligned}$$

of abelian monoids.

Remark 4.2. Let (R, P_R, α_R) be a pre-log ring, and M an R/p -module, and $m \in M$ an element. Then as in [Shi26, Prop 2.8], the map

$$\begin{aligned} \{\text{pre-log ring maps } s: R \rightarrow W_2(R, M, m) \text{ with } \text{res} \circ s = \text{id}\} &\longrightarrow \{\log \text{ FW-derivations } d: R \rightarrow M\} \\ s &\longmapsto \text{pr}_M \circ s \end{aligned}$$

is a bijection.

There is an extension of Example 2.1.

Example 4.3. Let (R, P_R, α_R) be a pre-log $\mathbb{F}_p$ -algebra. Then the arithmetic extension $W_2(R, R, 1)$ is given by the pre-log ring $(W_2(R), P_R \oplus R, \beta_R)$ where β_R is the map $P_R \oplus R \rightarrow W_2(R)$, $(a, x) \mapsto (\alpha_R(a), \alpha_R(a)^p x)$. By definition, this pre-log ring receives a map from the pre-log ring $(W_2(R), P_R, [\alpha_R(-)])$.

We have a pullback description of $(W_2(R, M, m), P_R \oplus M, \beta_R)$ as in Remark 2.10. For this, we have to construct the analogue $\psi_{(R, P_R, \alpha_R)}$ for animated pre-log $\mathbb{F}_p$ -algebras (R, P_R, α_R) .

Construction 4.4. Let (R, P_R, α_R) be an animated pre-log $\mathbb{F}_p$ -algebra. Then the animated ring $R \otimes_{\mathbb{Z}} \mathbb{F}_p$ is equivalent to the trivial square-zero extension $R \oplus R[1]$ (**not** as an animated $\mathbb{F}_p$ -algebra), thus it is the trivial square zero extension $(R \oplus R[1], P_R \oplus \Omega^\infty(R[1]))$ ^{4.1} of the pre-log ring (R, P_R, α_R) by the R -module $R[1]$, which can be viewed as an animated pre-log $\mathbb{F}_p$ -algebra (**no longer** a trivial square-zero extension as such), denoted by $(R \otimes_{\mathbb{Z}} \mathbb{F}_p, P_R \oplus \Omega^\infty(R[1]))$.

Construction 4.5. Let (R, P_R, α_R) be an animated pre-log $\mathbb{F}_p$ -algebra. Then the map $\psi_{(R, P_R, \alpha_R)}$ of animated pre-log rings is defined to be the composite map

$$(R, P_R, \alpha_R) \xrightarrow{\text{can}} (R \otimes_{\mathbb{Z}} \mathbb{F}_p, P_R \oplus \Omega^\infty(R[1])) \xrightarrow{\varphi_{(R \otimes_{\mathbb{Z}} \mathbb{F}_p, P_R \oplus \Omega^\infty(R[1]))}} (R \otimes_{\mathbb{Z}} \mathbb{F}_p, P_R \oplus \Omega^\infty(R[1]))$$

of animated pre-log rings, where the Frobenius map on $P_R \oplus \Omega^\infty(R[1])$ is multiplication by p . Note that the multiplication by p on $\Omega^\infty(R[1])$ is null-homotopy, and the null-homotopy is functorially given by the map $\mathbb{F}_p \rightarrow R$ of animated rings.

^{4.1}. We omit the map $P_R \oplus \Omega^\infty(R[1]) \rightarrow (\Omega^\infty(R \oplus R[1]), \cdot)$ in the notation.

Proposition 4.6. *Let (R, P_R, α_R) be a pre-log ring with $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$, and M an R/p -module, and $m \in M$ an element. Then the arithmetic extension $(W_2(R, M, m), P_R \oplus M, \beta)$ of (R, P_R, α_R) by M coincides with the pullback W_2^{pb} of the diagram*

$$\begin{array}{ccc} & (R, P_R, \alpha_R) & \\ & \downarrow \text{can} & \\ & (\bar{R}, P_R, \text{can} \circ \alpha_R) & \\ & \downarrow \psi_{(\bar{R}, P_R)} & \\ (\bar{R} \oplus \text{cofib}(m), P_R \oplus \Omega^\infty \text{cofib}(m)) & \longrightarrow & (\bar{R} \otimes \mathbb{F}_p, P_R \oplus \Omega^\infty(\bar{R}[1])) \end{array}$$

of pre-log rings, where the bottom left is the trivial square-zero extension of $(\bar{R}, P_R, \text{can} \circ \alpha_R)$ by $\text{cofib}(m)$, where $\text{cofib}(m)$ is viewed as a map $\bar{R} \rightarrow M$ of $\bar{R}$ -modules^{4.2}.

Proof. By Remark 2.11, the pullback W_2^{pb} has underlying ring $W_2(R, M, m)$. Moreover, the underlying abelian monoid is the pullback of the diagram

$$\begin{array}{ccc} & P_R & \\ & \downarrow & \\ P_R \oplus \Omega^\infty \text{cofib}(m) & \longrightarrow & P_R \oplus \Omega^\infty(\bar{R}[1]) \end{array}$$

of animated abelian monoids, where the vertical map is the composite map

$$P_R \xrightarrow{p} P_R \xrightarrow{\text{id}_{P_R}} P_R \oplus \Omega^\infty(\bar{R}[1]),$$

see Construction 4.5. It follows that the above pullback is the animated monoid $P_R \oplus \Omega^\infty M$, which is concentrated in degree 0. This shows that the pullback W_2^{pb} is a classical pre-log ring.

It remains to check that the exponential map $e^{\text{pb}}: P_R \oplus M \rightarrow (W_2(R, M, m), \cdot)$ in W_2^{pb} coincides with the map $\beta_R: P_R \oplus M \rightarrow (W_2(R, M, m), \cdot)$. Tracing through the identification in Remark 2.11 and Proposition 2.6, one can see that the map $e^{\text{pb}}|_{P_R}$ is the composite

$$P_R \xrightarrow{\alpha_R} (R, \cdot) \xrightarrow{[-]} (W_2(R, M, m), \cdot)$$

of abelian monoids^{4.3}, which is the same as $\beta_R|_{P_R}$. Now we identify $e^{\text{pb}}|_M$, which by definition, appears in the map

$$\begin{array}{ccccc} M & \xrightarrow{e^{\text{pb}}|_M} & \text{GL}_1(W_2(R, M, m)) & \xrightarrow{\text{res}} & \text{GL}_1(R) \\ \parallel & & \downarrow & & \downarrow \\ M & \xrightarrow{1+(-)} & \text{GL}_1(\bar{R} \oplus \text{cofib}(m)) & \longrightarrow & \text{GL}_1(\bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p) \end{array}$$

of fiber sequences in $D(\mathbb{Z})$ (and everything in question is actually connective). Comparing this with the map

$$\begin{array}{ccccc} M & \xrightarrow{V} & W_2(R, M, m) & \xrightarrow{\text{res}} & R \\ \parallel & & \downarrow & & \downarrow \\ M & \longrightarrow & \bar{R} \oplus \text{cofib}(m) & \longrightarrow & \bar{R} \otimes_{\mathbb{Z}} \mathbb{F}_p \end{array}$$

of fiber sequences in $D(\mathbb{Z})$ (see Remark 2.13), we see that $e^{\text{pb}}|_M = 1 + V = \beta_R|_M$. $\square$

^{4.2.} Since M is a $\pi_0(\bar{R})$ -module, it is also an $\bar{R}$ -module.

^{4.3.} It is essentially deduced from the fact that, for every animated ring R (or equivalently, every finitely generated polynomial ring R), the Teichmüller map $[-]: (\Omega^\infty R, \cdot) \rightarrow (\Omega^\infty W_2(R), \cdot)$ is given by the diagram

$$\begin{array}{ccc} (\Omega^\infty R, \cdot) & \xrightarrow{\text{id}} & (\Omega^\infty R, \cdot) \\ \downarrow p & & \downarrow \Omega^\infty \psi_R \\ (\Omega^\infty R, \cdot) & \xrightarrow{\Omega^\infty(\text{can})} & (\Omega^\infty(R \otimes_{\mathbb{Z}} \mathbb{F}_p), \cdot) \end{array}$$

of animated abelian monoids, where we realize $W_2(R)$ as the pullback $R \times_{\text{can}, R \otimes_{\mathbb{Z}} \mathbb{F}_p, \psi_R} R$.

Thus we extend the arithmetic extension and logarithmic Frobenius–Witt derivations to general animated pre-log rings.

Definition 4.7. Let (R, P_R, α_R) be an animated pre-log ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$, and M a connective $\bar{R}$ -module, and $m : \bar{R} \rightarrow M$ a map of $\bar{R}$ -modules. Then we define

- the arithmetic extension of animated pre-log ring of (R, P_R, α_R) by (M, m) to be the pullback of the diagram

$$\begin{array}{ccc} & (R, P_R, \alpha_R) & \\ & \downarrow \text{can} & \\ & (\bar{R}, P_R, \text{can} \circ \alpha_R) & \\ & \downarrow \psi_{(\bar{R}, P_R)} & \\ (\bar{R} \oplus \text{cofib}(m), P_R \oplus \Omega^\infty \text{cofib}(m)) & \longrightarrow & (\bar{R} \otimes_{\mathbb{F}_p}, P_R \oplus \Omega^\infty(\bar{R}[1])) \end{array}$$

of animated pre-log rings.

- the anima $\text{LFWDer}((R, P_R, \alpha_R), (M, m))$ of logarithmic Frobenius–Witt derivations is defined to be

$$\text{Hom}_{\text{LAlg}_{(R, P_R, \alpha_R)}^{\text{an}}}((R, P_R, \alpha_R), (W_2(R, M, m), P_R \oplus \Omega^\infty(\bar{R}[1])), \beta_R).$$

Arguing as in Theorem 3.6, we see that

Theorem 4.8. Let (R, P_R, α_R) be an animated pre-log ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$, and M a connective $\bar{R}$ -module, and $m : \bar{R} \rightarrow M$ a map of $\bar{R}$ -modules. Then the functor

$$\begin{array}{ccc} D_{\geq 0}(\bar{R})_{\bar{R}} & \longrightarrow & \text{An} \\ (M, m) & \longmapsto & \text{LFWDer}((R, P_R, \alpha_R), (M, m)) \end{array}$$

is co-represented by $(\text{fib}(\partial_{(R, P_R, \alpha_R)}), m_{(R, P_R, \alpha_R)})$, where $m_{(R, P_R, \alpha_R)}$ is the fiber of the map

$$\partial_{(R, P_R, \alpha_R)} : L_{(R, P_R, \alpha_R)/\mathbb{Z}} \otimes_{\bar{R}}^{\mathbb{L}} \bar{R} \longrightarrow \bar{R}[1]$$

of $\bar{R}$ -modules classifying the composite map

$$(R, P_R, \alpha_R) \xrightarrow{\text{can}} (\bar{R}, P_R, \text{can} \circ \alpha_R) \xrightarrow{\psi_{\bar{R}}} (\bar{R} \oplus \bar{R}[1], P_R \oplus \Omega^\infty(\bar{R}[1]))$$

of animated pre-log rings into the trivial square-zero extension of $(\bar{R}, P_R, \text{can} \circ \alpha_R)$ by $\bar{R}[1]$.

Similarly to Theorem 3.15 and Remark 3.16, we have

Theorem 4.9. Let (R, P_R, α_R) be an animated pre-log ring with $\bar{R} := R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$. Then the map $\partial_{(R, P_R, \alpha_R)}$ can be identified with the map

$$L_{(R, P_R, \alpha_R)/\mathbb{Z}} \otimes_{\bar{R}, \psi_R}^{\mathbb{L}} \bar{R} \longrightarrow L_{(\bar{R}, P_R, \text{can} \circ \alpha_R)/(R, P_R, \alpha_R)}$$

induced by the commutative diagram

$$\begin{array}{ccc} \mathbb{Z} & \longrightarrow & (R, P_R, \alpha_R) \\ \downarrow & & \downarrow \\ (R, P_R, \alpha_R) & \xrightarrow{\psi_R} & (\bar{R}, P_R, \text{can} \circ \alpha_R) \end{array}$$

of animated pre-log rings, as in Construction 3.22.

5 A regularity criterion

In this section, independently of previous developments, we give a regularity criterion of Noetherian rings without F -finiteness or locality (Theorem 5.2). This is inspired by [BM23, Lem 4.18].

Lemma 5.1. *Let $(R, \mathfrak{m}, k)$ be a local ring with residue field k of char p , with $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$.*

- *Suppose that the Frobenius–Witt cotangent complex FL_R is flat as an $\bar{R}$ -module. Then the cotangent complex $L_{k/R}$ is concentrated in degrees ≤ 1 .*
- *Conversely, suppose that the cotangent complex $L_{k/R}$ is concentrated in degrees ≤ 1 . Then the k -module $FL_R \otimes_{\bar{R}}^{\mathbb{L}} k$ is concentrated in degrees ≤ 0 .*

Proof. Let $\bar{k} := k \otimes_{\mathbb{Z}} \mathbb{F}_p$. We examine the transitivity sequence ([Shi26, Thm 5.3] or Construction 3.24)

$$FL_R \otimes_{\bar{R}}^{\mathbb{L}} \bar{k} \longrightarrow FL_k \longrightarrow L_{k/R} \otimes_{k, \psi_k}^{\mathbb{L}} \bar{k}$$

in $D(\bar{k})$. By [Shi26, Cor 6.4], the $\bar{k}$ -module FL_k is equivalent to the $\bar{k}$ -module $L_{k/\mathbb{F}_p} \otimes_{k, \psi_k}^{\mathbb{L}} \bar{k}$. We base change this fiber sequence along the map $\bar{k} = k \otimes_{\mathbb{Z}} \mathbb{F}_p \rightarrow k$ induced by id_k and the $\mathbb{F}_p$ -algebra structure on k , we get a fiber sequence

$$FL_R \otimes_{\bar{R}}^{\mathbb{L}} k \longrightarrow L_{k/\mathbb{F}_p} \otimes_{k, \varphi_k}^{\mathbb{L}} k \longrightarrow L_{k/R} \otimes_{k, \varphi_k}^{\mathbb{L}} k \quad (5.1)$$

in $D(k)$. Since the k -module $L_{k/\mathbb{F}_p}$ is actually flat [BLM18, Prop 9.5.1], so is the k -module $L_{k/\mathbb{F}_p} \otimes_{k, \varphi_k}^{\mathbb{L}} k$.

Now we suppose that the $\bar{R}$ -module FL_R is flat. Then the fiber sequence (5.1) implies that the k -module $L_{k/R} \otimes_{k, \varphi_k}^{\mathbb{L}} k$ has Tor-amplitude ≤ 1 . Since k is a field, the Frobenius map $\varphi_k: k \rightarrow k$ is faithfully flat. It follows that the k -module $L_{k/R}$ has Tor-amplitude ≤ 1 , or equivalently, is concentrated in degrees ≤ 1 .

Conversely, suppose that the cotangent complex $L_{k/R}$ is concentrated in degrees ≤ 1 , then so is the k -module $L_{k/R} \otimes_{k, \varphi_k}^{\mathbb{L}} k$ by flatness of $\varphi_k: k \rightarrow k$. It follows from the fiber sequence (5.1) that the k -module $FL_R \otimes_{\bar{R}}^{\mathbb{L}} k$ is concentrated in degrees ≤ 0 . $\square$

Theorem 5.2. *Let R be a Noetherian ring with $\bar{R} := R \otimes_{\mathbb{Z}} \mathbb{F}_p$.*

- *Suppose that the ring R is p -local (i.e. $p \in \text{Rad}(R)$), and the $\bar{R}$ -module FL_R is flat. Then the Noetherian ring R is regular.*
- *Conversely, suppose that the ring R is regular. Then the $\bar{R}$ -module FL_R is flat.*

Proof. For every prime ideal $\mathfrak{p} \in \text{Spec}(R/p)$, we will denote by $\kappa(\mathfrak{p})$ the residue field $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} = \text{Frac}(R/\mathfrak{p})$.

Suppose that the $\bar{R}$ -module FL_R is flat. Then for every maximal ideal $\mathfrak{m} \subseteq R$, since $p \in \text{Rad}(R) \subseteq \mathfrak{m}$ (i.e. $\mathfrak{m} \in \text{Spec}(R/p)$), the residue field $\kappa(\mathfrak{m}) = R/\mathfrak{m}$ is of char p . Note that the $\bar{R}_{\mathfrak{m}}$ -module $FL_{R_{\mathfrak{m}}}$, which is equivalent to the $\bar{R}_{\mathfrak{m}}$ -module $FL_R \otimes_{\bar{R}}^{\mathbb{L}} \bar{R}_{\mathfrak{m}}$ by transitivity sequence associated to $R \rightarrow R_{\mathfrak{m}}$ (cf. [Shi26, Rem 5.5]), is flat. It follows from Lemma 5.1 that the cotangent complex $L_{\kappa(\mathfrak{m})/R_{\mathfrak{m}}}$ is concentrated in degrees ≤ 1 , and by [Iye07, Prop 8.12] (where we use Noetherianness), the Noetherian local ring $R_{\mathfrak{m}}$ is regular. It follows from [Sta21, Tag 0AFS] that the Noetherian ring R is regular.

Conversely, suppose that the Noetherian ring R is regular. By [BLM18, Lem 9.5.2]^{5.1} and connectivity of FL_R , it suffices to check that, for every prime ideal $\mathfrak{p} \in \text{Spec}(R/p)$, the connective $\kappa(\mathfrak{p})$ -module $FL_R \otimes_{\bar{R}}^{\mathbb{L}} \kappa(\mathfrak{p})$ is concentrated in degrees ≤ 0 . This follows from applying Lemma 5.1 to the regular local ring $R_{\mathfrak{p}}$. $\square$

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^{5.1} Since for every animated ring R , by definition, a connective R -module M is flat if and only if the connective $\pi_0(R)$ -module $M \otimes_R^{\mathbb{L}} \pi_0(R)$ is flat. Thus [BLM18, Lem 9.5.2] extends to animated rings.

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